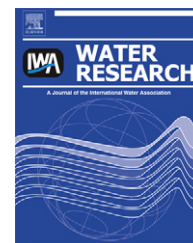


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Leaching of Br^- , metolachlor, alachlor, atrazine, deethylatrazine and deisopropylatrazine in clayey vadoze zone: A field scale experiment in north-east Greece

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ABSTRACT

An extensive four-year research program has been carried out to explore and acquire knowledge about the fundamental agricultural practices and processes affecting the mobility and bioavailability of pesticides in soils under semi-arid Mediterranean conditions. Pesticide leaching was studied under field conditions at five different depths using suction cups. Monitoring of metolachlor, alachlor, atrazine, deethylatrazine (DEA), deisopropylatrazine (DIA), and bromide ions in soil water, as well as dye patterns made apparent the significant role of preferential flow to the mobility of the studied compounds. Irrespective to their adsorption capacities and degradation rates, atrazine, metolachlor and bromide ions were simultaneously detected to 160 cm depth. Following 40 mm irrigation, just after their application, both alachlor and atrazine were leached to 160 cm depth within 18 h, giving maximum concentrations of 211 and 199 $\mu\text{g L}^{-1}$, respectively. Metolachlor was also detected in all depth when its application was followed by a rainfall event (50 mm) two weeks after its application. The greatest concentrations of atrazine, alachlor and metolachlor in soil water were 1795, 1166 and 845 $\mu\text{g L}^{-1}$, respectively. The greatest concentrations of atrazine's degradation products (both DEA and DIA) appeared later in the season compared to the parent compound. Metolachlor exhibited the greatest persistence with concentrations up to 10 $\mu\text{g L}^{-1}$ appearing in soil water 18 months after its application. Brilliant blue application followed by 40 mm irrigation clearly depict multi-branching network of preferential flow paths allowing the fast flow of the dye down to 150 cm within 24 h. This network was created by soil cracks caused by shrinking of dry soils, earthworms and plant roots. Chromatographic flow of the stained soil solution was evident only in the upper 10–15 cm of soil.

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1. Introduction

Soil applied herbicides are exposed to environmental conditions, and they can be degraded by the influence of physical,

chemical and biological factors, volatilized, adsorbed by soil colloids and transported offsite by surface runoff and leaching. The quantity of each pesticide that can be found in groundwater depends on the significance of all the above

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mentioned processes (Vryzas et al., 2007a; Arias-Estévez et al., 2008). Moreover, agricultural practices followed during and after pesticide application may also be the reason of intensifying pesticide leaching potential.

The herbicides atrazine (2-chloro-4-ethylamino-6-isopropylamino-s-triazine) alachlor [2-chloro-N(2,6-diethylphenyl)-N-(methoxy-methyl)acetamide] and metolachlor [2-chloro-N-(2-ethyl-6-methylphenyl)-N-(2-methoxy-1-methylethyl) acetamide] had been widely used for weed control in the corn fields of Greece and are still registered for use in other countries. The presence of these herbicides has frequently been detected in rain, surface- and groundwater worldwide (Charizopoulos and Papadopoulou-Mourkidou, 1999; Kolpin et al., 1998; Cerejeira et al., 2003; Konstantinou et al., 2006; Vryzas et al., 2009, 2011a). Contamination of groundwater of major corn producing areas of Greece by atrazine, deethylatrazine (2-amino-4-chloro-6-isopropylamino-s-triazine) (DEA), deisopropylatrazine (2-chloro-4-ethylamino-6-amino-s-triazine) (DIA) -two of the major atrazine's conversion products- alachlor and metolachlor was limited to the Ardas River valley in Northern Greece (Papastergiou and Papadopoulou-Mourkidou, 2001; Vryzas et al., 2011b).

Description of water flow and solute transport under field conditions is essential for evaluating the ground water contamination. Under chromatographic flow (uniform wetting front) conditions, a pesticide usually remains in the soil matrix for a sufficient time, allowing its adsorption and/or degradation. However, in compact clay soils, pesticides can be leached rapidly to groundwater via macropores, bypassing the majority of the soil matrix. Many researchers have reported rapid pesticide, tracer and colloid movement through preferential flow (Fortin et al., 2002; Burkhardt et al., 2008; Jarvis et al., 2008). Preferential flow in macropores has been included in dual-porosity, dual-permeability or multi porosity/permeability prediction models (Jarvis et al., 2009; Simunek et al., 2003; Gerke, 2006). This constitutes a major scientific challenge due to the complexity of macroporosity and its distribution throughout soil profile. The simulation of pesticide transport with models provides a cost-effective and rapid assessment tool. However, if these models are to be useful and accurate, their evaluation and comparison with data obtained from field scale experiments, especially under preferential flow conditions, is indispensable. In addition, according to Allaire et al. (2009), a standardization of an

approach to compare results from different methods (monitoring and modeling) should be developed. In the present study, during the four-year experimental period, 1605 soil water samples from five depths were analyzed and the raw data are provided for model calibration.

The paper first presents monitoring data of alachlor, metolachlor, atrazine, DEA, DIA and bromide ions in soil water samples taken from five different depths in corn fields during four growing seasons, which were managed under local agricultural practices. We then compare leaching behavior of the studied compounds, by examining the relationships between their concentration ratios, to improve our understanding of the transport of pesticides with different adsorption and degradation profile. The contribution of preferential flow to the overall transport of various chemicals in relation to the followed agricultural practice was accessed with a view to promote changes of minimizing groundwater contamination under semi-arid Mediterranean conditions.

2. Materials and methods

2.1. Experimental fields

Three experimental field sites were selected in north-eastern Greece (41°38'1.48" N, 26°20'6.74"E; 41°36'14.51"N, 26°20'48.27"E; 41°37'8.26"N, 26°25'4.11"E), close to the Greek/Bulgarian/Turkish borders, in order to investigate the leaching potential of corn-applied herbicides (see Fig. S1 in the Supporting Information). An agreement was made with the landowners to continue growing corn for the four-year experimental period. Management decisions and agricultural practices were decided by the landowners and were typical of corn farming in Northern Greece (Table 1). Drinking water wells, where herbicide residues have been previously found, were located adjacent to the fields (Papastergiou and Papadopoulou-Mourkidou, 2001). These wells and boreholes installed at the edge of each field were monthly monitored during the experiment. Intact soil cores were removed from each field, and each core was divided into five layers (0–10, 10–20, 20–40, 40–80, 80–110 cm depth) to characterize the soil textural properties (Sheldrick and Wang, 1993), total organic carbon content (Nelson and Sommers, 1982) and pH (McLean, 1982).

Table 1 – Summary of the agricultural practice applied in the experimental fields.

Treatment	Information	Experimental periods			
		1st	2nd	3rd	4th
Tillage	plough 35 cm/harrow 10–15 cm	Nov./Feb.	Oct./Feb	Nov./Jan	Dec./Feb
Fertilizer	(22-11-0) or (27-9-0 + 4S + 1.5 Zn) (650–700) Kg ha ⁻¹	20–30 Mar.	15–20 Mar.	5-10 Apr.	15-25 Apr.
Seeding	65000–70000 plants ha ⁻¹ 17 × 75 cm	10 Apr.	30 Mar.	20 Apr.	4 May
Herbicide application	Alachlor or metolachlor and KBr	10 Apr.	30 Mar.	20 Apr.	4 May
	Atrazine	10 Apr.	7 May	15 May	22 May
	Through corn rows at 10–15 cm depth	20–25 May	20–25 May	25–30 May	1–5 June
Cultivation	Through corn rows at 10–15 cm depth	20–25 May	20–25 May	25–30 May	1–5 June
Fertilizer	(26-0-0 + Ca) (500–550) Kg ha ⁻¹	10–15 June			
Start of irrigation	Sprinkler irrigation Every 8–10 days	10–15 June			
Stop of irrigation		16 Aug.	26 Aug.	25 Aug.	10 Aug.
Harvest	2–10 Oct.				

2.2. Suction cup installation and soil water sampling

Three clusters of suction cups were installed in each field (see Fig. S2 in the Supporting Information). Each cluster consisted of five suction cups placed at five different depths. Two types of suction cups were used; the 1900 soil water sampler and the 1922 ultra water sampler (Soilmoisture Equipment Corp., Santa Barbara, USA). The 1922 samplers are entirely made of 1 bar high-flow ceramic material, it has a length of 25 cm, and collects soil water from the upper 0–25 cm soil layer. The 1900 samplers of four sizes (of 35-, 60-, 100- and 160-cm length) were respectively installed at 35, 60, 100 and 160 cm below ground surface (bgs) to collect soil water of the specific depth (see Supplementary data for additional details). In a total of 58 sampling events 1605 soil water samples were collected from all sampling depths and were analyzed for the determination of pesticide residues.

2.3. Herbicide and tracer applications

The typical local agricultural practice concerning herbicide application in the studied area was followed during the first year of the experiment (worst case scenario). According to this practice, herbicides that control broadleaf and grass weeds were applied simultaneously and at high doses. Namely, the commercial formulation Lasso-AT EC (alachlor 33.6% and atrazine 14.4%) was applied at 6500 ml ha⁻¹ in field 2 followed by irrigation, while 7500 ml ha⁻¹ Primextra FW (metolachlor 30% and atrazine 20%) were applied on fields 1 and 3 followed by mechanical incorporation. Irrigation or mechanical incorporation after herbicide application is common practice in the Ardas Valley area in order to prevent losses due to volatilization or photodegradation. In the second, third and fourth year of the experiment, atrazine (Gesaprim 500 FW) and metolachlor (Dual 96 EC) were applied separately in all fields at the lowest recommended doses (2000 and 1500 ml ha⁻¹ for Gesaprim and Dual, respectively) in order to investigate the safe use of those herbicides by minimizing the factors that intensify their leaching. From the second to the fourth year, metolachlor was applied as pre-emergence, while atrazine as an early postemergence herbicide. All commercial formulations and their spraying solutions were analyzed to determine their exact composition. The variability in concentration between each herbicide application was determined by analyzing, five surface soil samples (0–10 cm) from each field, collected just after each herbicide application and twelve paper filters (Whatman No. 2) which had been randomly positioned in each field before each herbicide application. Commercial formulations, their spraying solutions, soil samples and filter papers were analyzed according to the methods described by Vryzas and Papadopoulou-Mourkidou (2002) and Vryzas et al. (2007b).

In each growing season and prior to the first herbicide application, KBr was applied (22 g per m²) as a tracer for water on all fields, while Erioglaucine A (brilliant blue FCF, Fluca, Switzerland) was applied on field 2 in the fourth year (125 g per m²). Erioglaucine was applied on a 1 m² area and at a 5 m distance from a cluster of suction cups. Its application was followed by 40 mm irrigation (according to the local agricultural practice). Twenty four hours later, a trench was dug and

several vertical sections (2.5–5 cm thick and 150 cm deep) were excavated. Photos of the stained sections were taken prior to excavation of each slice.

2.4. Apparatus and respective operational parameters

Separation and quantification of target solutes were made according to the method described by Vryzas et al. (2009). The limits of detection (LODs) of all solutes ranged from 0.001 to 0.01 µg L⁻¹, and the limits of quantification (LOQ) ranged from 0.005 to 0.05 µg L⁻¹. Recoveries were periodically checked at 0.01, 0.05, 0.1, 5, 100 µg L⁻¹ levels and were found to be greater than 76% in all cases.

2.5. Bromide and herbicide analyses

Bromide concentration was determined to a 0.5–1 ml sub-sample extracted from each soil water sampler using a bromide electrode model 96-36 ion plus (thermo Orion, Sedgeway Business Park Witchford, England). Selected soil water sub-samples were also analyzed using high performance liquid chromatography (HPLC) in order to confirm the results from the bromide electrode.

The remaining soil water sample was used for the pesticide residue analysis as described by Vryzas et al. (2009).

3. Results and discussion

Details concerning the spatial variability of the herbicides applied and rainfall/irrigation water can be found in the Supplementary data. Most layers in all fields were characterized as clay-loam or clay soils (see Table S1 in the Supplementary data). Shrinkage cracks were observed at the first soil layers after each dry period, and permanent macropores resulting from earthworm activity and/or root decay were found in the 0–160 cm soil layers during the installation of suction cups.

3.1. Alachlor

Alachlor was applied to all fields one year before the beginning of the experiment and was constantly detected in soil water samples of all depths in fields 1 and 3 at concentrations up to 14.2 µg L⁻¹ due to these previous applications. However the last detection of alachlor in fields 1 and 3 was 0.53 µg L⁻¹ and was observed on July tenth, 39 months after its application (data not shown).

In field 2, where alachlor was applied in the first year with atrazine in the form of the commercial formulation Lasso-AT, it was detected in all depths at concentrations up to 4.8, 0.6 and 0.2 µg L⁻¹ one, two and three years after its application, respectively. Alachlor was detected even in the last sampling, which was 40 months after its application, at concentrations greater than 0.1 µg L⁻¹ in a soil water sample from 60 cm depth (Fig. 1). Alachlor concentrations which were detected in soil water at all depths of field 2 consistent with those from the other two fields, indicating great persistence of the low alachlor concentrations in the soil water samples. To our knowledge, this is the first report of such extended alachlor

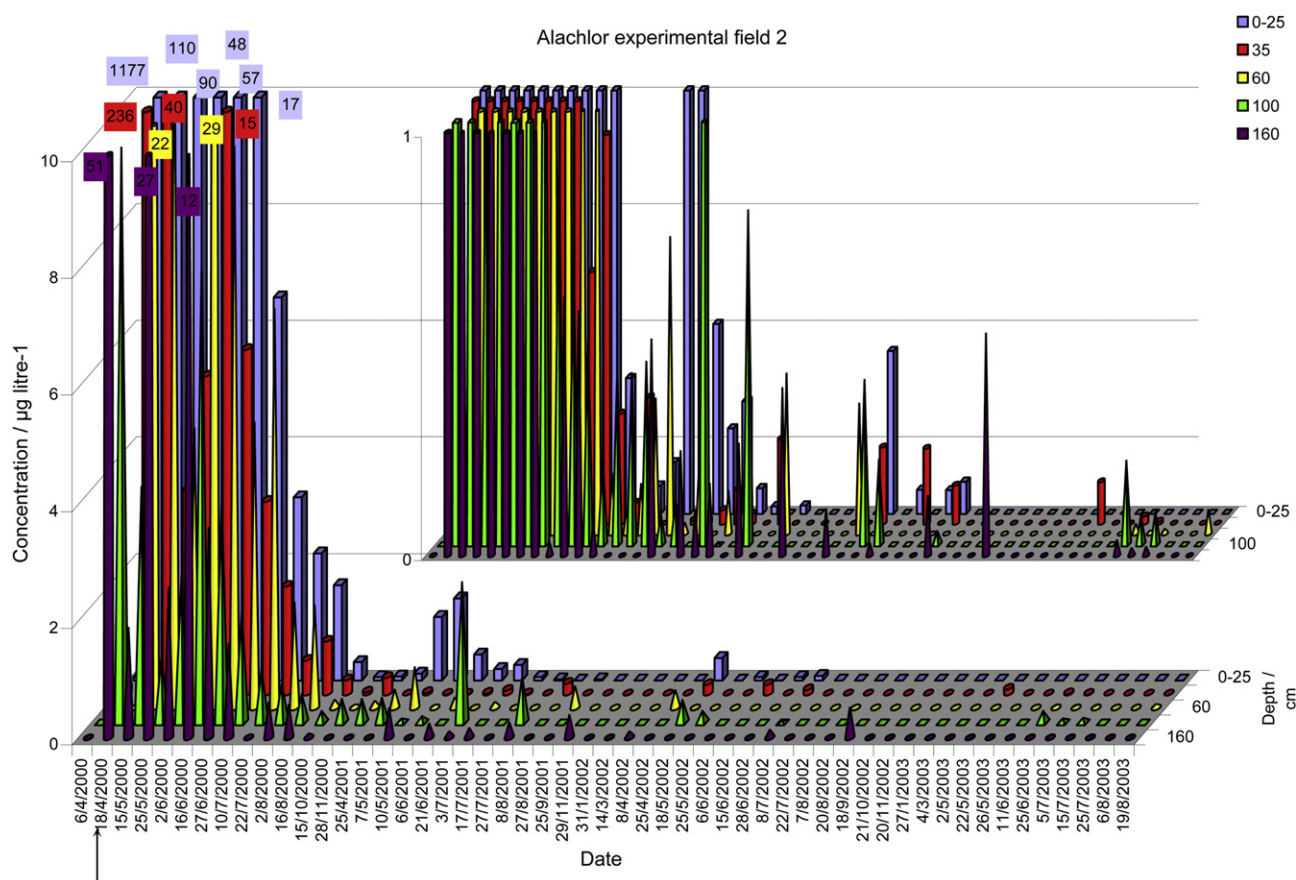


Fig. 1 – Mean alachlor concentrations ($\mu\text{g L}^{-1}$) in soil water samples from five different depths in experimental field No. 2 (alachlor was applied only to field No. 2, the arrow points at the date of application, Date as day/month/year).

detection under field conditions. In most previous studies, alachlor had shown a persistence ranging from 4 to 14 weeks while its DT_{50} value ranged from 1 to 40 days (Weed et al., 1995; Walker et al., 1992; Helling et al., 1988; Wienhold and Gish, 1994; Walker and Welch, 1991; Jones et al., 1990). In addition, according to Weed et al. (1998), alachlor DT_{90} ranged from 17 to 30 days. According to the pesticide properties database (Agriculture & Environment Research Unit (AERU), 2011) alachlor is slightly mobile with K_f values ranged from 11.8 to 20.5 ml/g and non-persistent with DT_{50} values ranged from 14 to 35 days. However, in the same source DT_{50} values up to 80 days have been reported. The lower limits of detection in soil water analyses compared to the soil analyses (soil water analyses being 3–20 times more sensitive than soil analyses) might have been the reason for the detection of alachlor 39–40 months after its application.

The leaching behavior of alachlor in the soil water of field No. 2 after its application in the first year is shown in Fig. 1. In April, during the first sampling after its application, alachlor was detected at great concentrations in all depths. Higher alachlor concentrations were detected in soil water samples from the 0–25- and 35-cm than in 100- and 160-cm depth.

In the soil water samples from the first depth (0–25 cm), alachlor concentrations rapidly decreased between the first and second sampling and afterward gradually with no fluctuations during the first growing season. High alachlor

concentrations were observed in soil water samples from 0 to 25 cm at the beginning of the next growing seasons (2nd and 3rd) although alachlor had not been applied in those years, indicating intense desorption phenomena of alachlor after great rainfall or irrigation events.

In the soil water samples from 35-, 60-, 100- and 160-cm depths alachlor concentrations fluctuated throughout 2000, showing multiple peak concentrations. It is worth noticing that the detection of great alachlor concentrations in samples collected from deep layer coincided with the detection of lower concentrations in shallower samples, while in the next sampling (next irrigation or rainfall event), the opposite happened. This is an evident that alachlor is not transported uniformly in the soil profile but its concentration is fluctuated with the soil depth. The local agricultural practice of 30–40 mm irrigation after herbicide application seems to be a crucial factor for the movement of herbicides through preferential flow paths to different depths.

From these first results at the beginning of the experiment, the significant role of preferential flow in the leaching of the herbicides to groundwater was evident. These results were in accordance with other researchers; Funari et al. (1998) reported alachlor detection at concentrations up to 107 $\mu\text{g L}^{-1}$ in soil water samples from 90 cm bgs. In addition, Laabs et al. (2002) detected alachlor as well as 7 more pesticides in deep soil water samples earlier in clay than in sandy soils, and

finally Jaynes et al. (2001) detected alachlor at $10 \mu\text{g L}^{-1}$ in soil water samples from 120 cm bgs within 2 h of an irrigation event.

3.2. Metolachlor

Metolachlor was applied in all fields in all growing seasons except for field No. 2 in the first year. However, it was detected in field No. 2 during that period due to previous years' applications; its concentrations were up to $10 \mu\text{g L}^{-1}$ with the greatest detected in soil water samples from 0 to 25 cm.

Metolachlor was detected in 95% of the samples that were collected in first year from field No. 2, although it had been applied a year before, while alachlor was detected in 74% of the total soil water samples in the same field one year after alachlor application in second year. The difference between metolachlor and alachlor detection frequencies may be attributed to the greater persistence of metolachlor. In parallel dissipation studies that were conducted in soils from the same fields, the DT_{50} values of metolachlor ranged from 48 to 162 days depending on the sampling depth (Vryzas et al., 2011b). Other researchers had also found that alachlor showed lower persistence than that of metolachlor (Sanyal and Kulshrestha, 1999; Hoyle and Arthur, 2000).

The leaching behavior of metolachlor in the soil water of all fields after repeated annual applications is shown in Fig. 2. The greatest metolachlor concentration was $845 \mu\text{g L}^{-1}$ and was detected in soil water samples from the first depth (0–25 cm) of field No. 1. In first year, when metolachlor was applied in combination with atrazine in the form of the formulation product Primextra in fields No. 1 and 3, metolachlor was not detected at a depth below the 25-cm during first sampling, which was the case for alachlor as mentioned above. This could be attributed to the incorporation of the pesticide and the lack of irrigation after metolachlor application in fields No. 1 and 3. During the second sampling, one month after application, metolachlor was detected in great concentrations in soil water samples from all depths, but the greatest concentrations of the specific growing season for each depth were detected in the later samplings.

During the second year, the first soil water sample from all fields was collected one month after metolachlor application due to the lack of rainfall or irrigation. The greatest metolachlor concentrations in soil water samples from all depths were detected during the first sampling and gradually decreased afterward.

During the third year, the first sampling was conducted two weeks after metolachlor application. The greatest

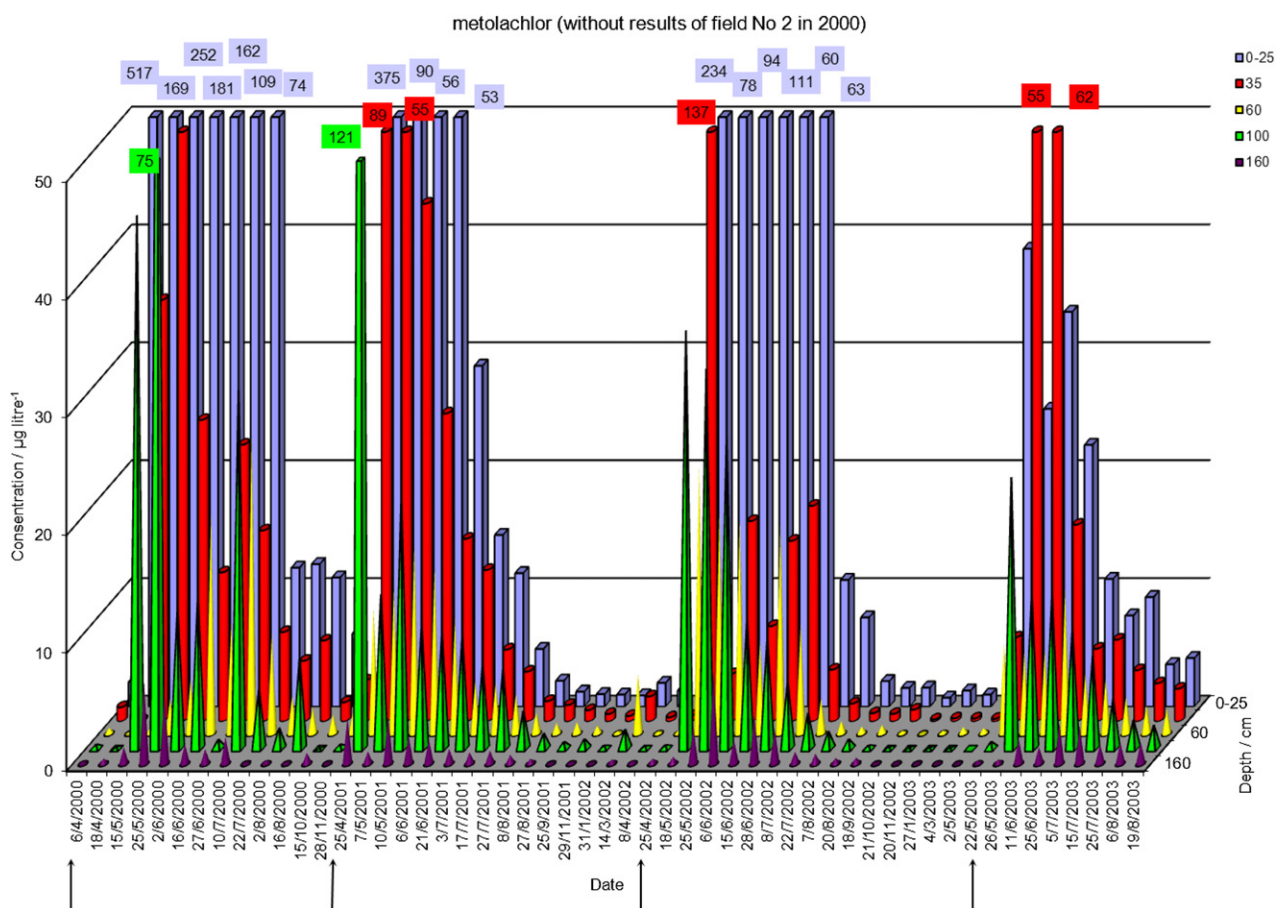


Fig. 2 – Mean metolachlor concentrations ($\mu\text{g L}^{-1}$) in soil water samples from five different depths in experimental fields (metolachlor was applied in all fields throughout the experimental period apart from field No2 in 2000, the arrows point at the dates of application, Date as day/month/year).

concentrations in soil water samples from 35-, 60-, 100-cm were detected in the first sampling after application, while the greatest concentrations in 160 cm were detected 3 weeks later.

Finally, during the fourth year, the level of metolachlor concentrations in soil water samples from all depths were lower than those of the previous years. In that year, the greatest metolachlor concentrations in soil water samples from 35 cm depth were detected in the third sampling after metolachlor application. The low metolachlor concentrations seem to be the result of the later metolachlor application. In the fourth year, metolachlor was applied later than in the three previous years, and its persistence could have been reduced due to the higher temperatures (3–5 °C greater than the previous years). Moreover, metolachlor application took place after spring raining season resulting in the reduction of preferential flow.

Although metolachlor has been detected both in surface- and groundwater worldwide and there are several studies concerning leaching of metolachlor, most of them have been conducted under different climatic conditions, soil characteristics and agricultural practices. Laabs et al. (2002) reported metolachlor concentrations up to 47 $\mu\text{g L}^{-1}$ in soil water samples from 35 cm bgs. Funari et al. (1998) detected up to 24 $\mu\text{g L}^{-1}$ metolachlor in drainage water (90 cm deep) within one month after its application. Novak et al. (2003) reported that macroporosity (in clay soils) and adsorption were the predominant factors influencing the leaching behavior of

metolachlor to drainage water. The soil properties and especially the distribution of macroporosity in combination with agricultural practice and weather differences seem to be the main reasons for the differences among studies. Smith and Parrish (1993) reported that metolachlor was not leached below 30 cm within 111 days after its application. Finally, Francaviglia and Capri, (2000) found that metolachlor concentrations in leachate of a 1.5 m lysimeter ranged from 1.63 to 22.87 $\mu\text{g L}^{-1}$ within 63–133 days after its application.

3.3. Atrazine

The above mentioned general observations concerning leaching of alachlor and metolachlor were also observed in the atrazine case. Fig. 3 presents the behavior of atrazine in the soil of all fields after repeated annual applications. In the first year, when atrazine was applied simultaneously with alachlor in field 2 followed by a 40-mm irrigation, both compounds were detected at concentrations up to 211 and 199 $\mu\text{g L}^{-1}$, respectively, in soil water samples from 160 cm bgs. The simultaneous detection of atrazine and alachlor in high concentrations at large depths only a few hours after their application indicates that dissipation and adsorption in that short time do not reduce the pesticide concentration in soil water. Under such conditions, pesticides with a different adsorption capacities and degradation rates could leach rapidly to deeper soil layers, where adsorption is less and

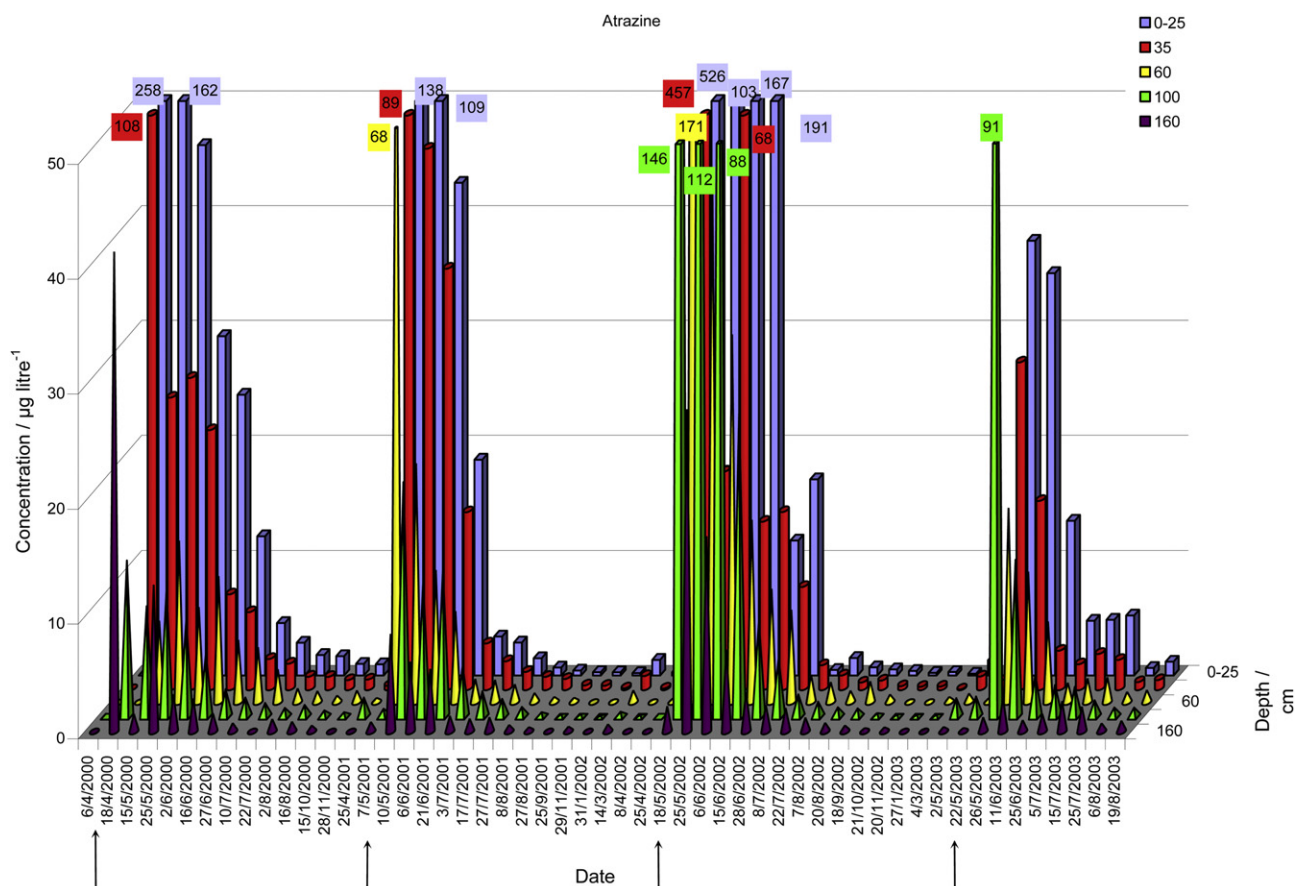


Fig. 3 – Mean atrazine concentrations ($\mu\text{g L}^{-1}$) in soil water samples from five different depths (atrazine was applied in all fields throughout the experimental period, the arrows point at the dates of application, Date as day/month/year).

degradation rates slower than those observed in surface soil layer. The detection of greater concentrations of alachlor compared to atrazine in all depths could also be attributed to the greater proportion of alachlor (2.3 times greater) in the Lasso-AT formulation. McGrath et al. (2010), in a modeling study, also suggested that the timing and frequency of rapid flow events may provide a useful measure of the rapid transport potential of pesticides. Also, as mentioned in the same study, there is a crucial need to acquire monitoring data in many parts of the world and under different conditions in order to validate the predictions derived from modelling studies.

The greatest atrazine concentrations in soil water samples from all depths were detected during the third year. In this year, rainfall followed atrazine application, and soil water samples were collected two days after the application. The greatest atrazine concentration ($1166 \mu\text{g L}^{-1}$) was detected in a soil water sample from 35 cm bgs. During the fourth year, the lowest concentrations of atrazine in all depths were detected, which was also the case for metolachlor.

Interesting observations were made by comparing the metolachlor to atrazine ($C_{\text{met}}/C_{\text{atr}}$) and the alachlor to atrazine ($C_{\text{al}}/C_{\text{atr}}$) concentration ratios during the first year when pesticide mixtures were used. In fields 1 and 3, where metolachlor and atrazine were applied simultaneously in first year, the concentration ratio ($C_{\text{met}}/C_{\text{atr}}$) in soil water samples from the 0–25 cm during the 1st and 2nd sampling ranged from 1.3 to 1.7 while the $C_{\text{met}}/C_{\text{atr}}$ in the applied formulation Primextra, was 1.5. That ratio gradually increased to 5 one month after the application and up to 13.3 two months later (see Fig. S3 in the Supplementary data). Three months after the Primextra application, the $C_{\text{met}}/C_{\text{atr}}$ ratio began to decrease and reached 2.9 seven months after the application. The low $C_{\text{met}}/C_{\text{atr}}$ ratio values, close to the Primextra formulation ratio, in the surface soil water samples (0–25 cm) shortly after the application indicate that atrazine and metolachlor have similar adsorption patterns in surface soil. The above mentioned observation was also confirmed by batch equilibrium adsorption studies in soil samples of the same fields (Vryzas et al., 2007a). According to Vryzas et al. (2007a), the K_d values of atrazine in surface soil samples of the experimental fields ranged from 0.88 to 2.38, while metolachlor exhibited slightly greater adsorption with K_d values ranged from 0.93 to 2.85 in the soil samples of the first 0–20 cm depth. The great $C_{\text{met}}/C_{\text{atr}}$ ratio values one and two months after the application could be attributed to the greater metolachlor persistence compared to that of atrazine or to the differences in their degradation pattern (Vryzas et al., 2011b). According to Vryzas et al. (2011b) the DT_{50} values of atrazine, in soil samples from the same experimental fields of the present study, ranged from 5 to 18 days in the surface layers and increased as the soil depth increased reaching the values of 43 days in the 80–110 cm depth layer, while metolachlor degraded at slower rates compared to atrazine with the respective DT_{50} values ranging from 56 to 72 days in surface soils and from 165 to 186 days in subsoils.

The $C_{\text{met}}/C_{\text{atr}}$ ratio in deeper soil water samples (25–160 cm) was lower than 1 during the first month after application, indicating a faster atrazine movement through preferential flow. Later the $C_{\text{met}}/C_{\text{atr}}$ ratio increased and was kept greater than 1 up to seven months after the application. The

percentage of the amount of atrazine that was detected in soil water samples from 25 to 160 cm depth during the first month after the application ranged from 63 to 87% of the total in those depths, while the respective percentage for metolachlor ranged from 53 to 57%, indicating that the metolachlor leaching period was more extended than that of atrazine (data are not shown).

In field 2 where alachlor and atrazine were applied simultaneously in first year, the concentration ratios $C_{\text{al}}/C_{\text{atr}}$ in soil water samples from the 0–25 cm during the 1st and 2nd sampling ranged from 1.7 to 2.1, while the $C_{\text{al}}/C_{\text{atr}}$ in the Lasso-AT formulation during the application was 2.3. This indicates a similar alachlor and atrazine adsorption behavior in the 0–25 cm soil layer. Then the $C_{\text{al}}/C_{\text{atr}}$ ratio gradually increased up to 3.8, and that could be attributed to the very high atrazine initial dissipation rate ($5 < DT_{50} < 18$ days) in the specific soils due to microbial adaptation as confirmed in laboratory experiments (Vryzas et al., 2011b). Afterward, the $C_{\text{al}}/C_{\text{atr}}$ ratio gradually decreased to 0.9 four months after the application, indicating that in spite of the initial high rate of atrazine dissipation, four months after their application alachlor was less persistent than atrazine (see Fig. S3 in the Supplementary data).

During first year in soil water samples from 25 to 160 cm depth of field No. 2, the $C_{\text{al}}/C_{\text{atr}}$ ratio in the 1st sampling after application was lower than 1. Later it had values greater than 1 up to 3 months after the application and finally decreased below 1, three months after the application (data are not shown).

Taking the $C_{\text{met}}/C_{\text{atr}}$ and $C_{\text{al}}/C_{\text{atr}}$ ratios at all depths into consideration, we could conclude that: a) metolachlor was more persistent than atrazine, b) atrazine initially leached more quickly than alachlor and metolachlor, c) atrazine showed a greater initial dissipation rate than alachlor but a greater amount of atrazine than that of alachlor was bio-available three to four months after the application. The above mentioned observations showed that adsorption, persistence and leaching are concentration dependent and interact with each other. Persistence and leaching of each pesticide depended on the pace of bound residue formation. Therefore, more sophisticated studies were conducted and the above general observations were confirmed (Vryzas et al., 2007a, 2011b).

Most studies concerning atrazine leaching had been conducted by analyzing soil samples or drainage water. Due to the sampling strategy of collecting soil water samples after a rainfall or irrigation event that was followed in this study, our results correspond better to studies focused on drainflow than soil studies. Mbuya et al. (2001) in contrast with our results found that atrazine concentrations in soil below 60 cm was lower than $5 \mu\text{g kg}^{-1}$, indicating low atrazine leaching potential compared with bromide ions. Southwick et al. (1995), who focused on the role of preferential flow in atrazine leaching reported that 95% of atrazine found in drainage water was detected during the first 21 days after application. Southwick et al. (1995) reported that the greater atrazine concentration in drainage water at 100 cm was detected in the first day after atrazine application. Our results are in agreement with those of Patterson et al. (2000) and Closea et al. (1998), who reported faster atrazine leaching from clay-loamy soils than from sandy soils.

Tasli et al. (1996), using the same application rate, found that atrazine concentration in soil water samples from 30, 50, 80 cm ranged from 0.3 to 15 $\mu\text{g L}^{-1}$, quite lower values than those we have detected especially during the early samplings, indicating greater preferential flow in our study. Also, in our study atrazine concentrations detected in all depths decreased rapidly within the first month after application, while Tasli et al. (1996) reported a more gradual decrease of atrazine's concentrations.

3.4. DEA and DIA

Metolachlor, atrazine, DEA and DIA were detected in 99, 95, 91 and 67% of the total soil water samples, respectively. Atrazine, DEA and DIA concentrations that were detected from the last irrigation (at the end of August) until the next year's application were lower than 5 $\mu\text{g L}^{-1}$, with DEA having the greatest values. In 92.5% of the total soil water samples where DEA was detected, it was showing greater concentrations than DIA with a $C_{\text{DEA}}/C_{\text{DIA}}$ ratio up to 33.

The concentrations of DEA and DIA in the soil water of all fields after repeated annual atrazine applications are shown in Figs. S4 and S5 in the Supplementary data, respectively. In contrast to the alachlor, metolachlor and atrazine cases, DEA and DIA concentrations gradually increased after atrazine application in soil water samples from 0 to 25 cm. The greatest DEA and DIA concentrations in all depths were not detected during the first sampling, as in the alachlor, metolachlor and atrazine cases, but much later. DEA and DIA concentrations in soil water samples from the first depth (0–25 cm) revealed lower variation among the growing seasons compared with the parent compound. The greatest DEA concentrations in soil water samples from the 0–25 cm were detected during the fourth year, when atrazine displayed the lowest concentrations compared to the other years, a characteristic incident of the favorable metabolic conditions prevailing due to the late atrazine application.

The concentrations of atrazine and DEA were examined in more detail to possibly identify processes controlling leaching. The DEA to atrazine ratio (DAR) has been used by Adams and Thurman (1991) to distinguish point source from nonpoint source contamination of groundwater. Although in recent studies (Vryzas et al., 2011b) it was found that the major atrazine conversion product in those fields was the hydroxyatrazine (HA), the DAR could also be used to explore atrazine leaching behavior because atrazine shows a closer adsorption capacity to DEA than to HA (Vryzas et al., 2007a). Variations of DAR values in 0–25, 100 and 160 cm bgs during the experimental period are shown in Fig. S6 in Supplementary data.

The DAR index was lower than 1 in soil water samples from 0 to 25 cm depth during the first year. In all the subsequent years and from the application date until July of each year, the DAR index in that depth was lower than 1 and then increased to greater than one until the next year's application.

The DAR increasing tendency during the growing season and the sharp decline just after the atrazine application were also observed in soil water samples from the 25–160 cm depths. In soil water samples from 35 cm depth the DAR index was greater than 1 only during the fourth year. The greatest

DAR value (7.9) was observed in winter of the third year in soil water samples from the 60 cm depth. In the same period, great DAR values in soil water samples from 100 to 160 cm were also observed.

As mentioned elsewhere (Vryzas et al., 2007a), atrazine showed a greater adsorption capacity than DEA in the specific soil samples; thus DEA leached more rapid than atrazine through chromatographic or preferential flow. The low DAR values just after the atrazine application were due to the early spring rainfall events resulting in fast atrazine preferential flow. The rainfall or irrigation just after atrazine application did not allow atrazine to remain in the biologically active upper soil layer for a sufficient period and be metabolized to DEA, which afterward could leach more rapid than atrazine due to its lower adsorption capacity. Our results are in agreement with those of Spalding et al. (2003a,b) who found similar DAR values in shallow groundwaters, and soil waters respectively.

3.5. Bromide ions

Bromide concentrations in soil water samples in all depths were at a mg L^{-1} range, approximately 1000 times greater than the corresponding alachlor, metolachlor and atrazine concentrations due to the greater application rates of KBr. High bromide concentrations were detected in all depths simultaneously or earlier than all other compounds tested.

The leaching behavior of bromide in the soil of all fields after repeated annual KBr applications are shown in Fig. S7 in Supplementary data. The greatest bromide concentrations in soil water samples from 0 to 25, 35 and 60 cm were detected in the first sampling of each growing season, verifying the significant role of the preferential flow in the specific area. When bromide application was followed by irrigation or rainfall, it was detected in all depths as in alachlor, metolachlor and atrazine cases. Bromide is a non-reactive tracer, which was not adsorbed by the soil colloids (Kidmose et al., 2010). The greatest bromide concentrations in soil water samples from 100 to 160 cm were not always detected in the first sampling after application, probably because the leaching mechanism of the non-adsorbing, non-degrading bromide tracer was not only dependent on the preferential but also on the chromatographic flow.

3.6. Brilliant blue FCF (eriglausine A)

The role of preferential flow in the leaching of alachlor, metolachlor, atrazine, DEA, DIA and bromide ions was visualized by tracing the preferential flow paths. Images of flow patterns typical of each type of preferential paths are presented in Figs. S8–S11 (see in the Supplementary data). The chromatographic flow was restricted in the first 10–15 cm depth, where the dye coverage was almost 80–100% (Fig. S8b Supplementary data). This observation corroborates previous findings by Vanderborght et al. (2002) who reported that the homogeneous flow pattern (chromatographic flow) was observed in the 0–20 surface soil, while below that depth the tracer followed an arborescent preferential network.

Up to 15 cm, most micropores were active (stained), while below that depth, preferential flow was obviously the main mechanism of dye movement. From the inspection of the dye

images, it was noted that there were different kinds and origins of preferential flow paths.

The most basic and widest preferential flow paths were due to the shrinkage of the clay minerals. These flow paths were relatively wide (0.5–5 cm) in the upper vadoze zone (see Fig. S8c, d in the Supplementary data), decreasing to much more narrow paths with the respective diameters ranging from 0.5 to 1 mm, below 150 cm depth (see Fig. S9a, b, c, d in the Supplementary data). Therefore, water and solutes transported through such a preferential network must have been traveling through a decreasing transport volume with an increasing velocity despite water uptake by the dry soil matrix.

Numerous preferential flow paths were also formed by corn and weed roots (Fig. S10a, b, c, d see in the Supplementary data). These paths were of a few millimeters width with roots absent or present in various stages of decay. Finally, earthworms forming burrows had a great contribution to the total preferential flow network existing in the vadoze zone of this area (see Fig. S11a, b, c, d in the Supplementary data). As a result of the existence of this network in the vadoze zone, erioglaucine was transported below the depth of 150 cm within 24 h following irrigation.

The dye movement was not perpendicular to ground surface but it took 100 cm lateral flow from its application point to reach the depth of 150 cm. The lateral movement was visualized in Fig. S8c, d (see in the Supplementary data). The fact that dye was found in deeper layers while it was absent in shallower layers indicates that it had moved horizontally.

The total distribution of the dyed macroporosity did not assume any regular arborescent or neural shape. It consisted of thick fissures through which the dye moved, and then the dye mass transferred from the preferential flow path into the soil matrix of the adjacent microporosity. In deeper layers, there was a dense preferential flow path network looking like a mosaic (see Fig. S9, S10 in the Supplementary data).

The surface soil layer (0–20 cm) had the role of a dye reservoir which supplied the preferential flow paths with dye after an intense rainfall or irrigation. Under Mediterranean semi-arid conditions prevailed during the growing season the preferential flow was intensified when rainfall or irrigation followed a drought period, because the clay minerals had shrunk resulting in the formation of denser network of preferential flow paths. As also mentioned by Hardie et al. (2011), antecedent soil moisture strongly influenced the type, depth and rate of dye tracer movement.

Our results are in agreement with those of Fortin et al. (2002) who have found that atrazine, metolachlor and bromides, inspite of their differences concerning their physicochemical properties- were simultaneously detected in drainage water at a depth of 120 cm. Jaynes et al. (2001) have also detected atrazine, alachlor and bromide at a depth of 120 cm 2 h after irrigation (8 mm).

Kosmas et al. (1991) reported rapid water movement through preferential flow at the beginning of irrigation, while the water flow through preferential flow paths was reduced during the irrigation because as macroporosity became saturated, the moist clay minerals swelled and the total macroporosity volume was reduced. Kung et al. (2000), however, reported that tracers moved more quickly through

preferential flow during the irrigation as the soil became moist and most pores became hydraulically active.

Our results corroborate the above mentioned ostensibly contradictory findings, as we observed that in dry shrunk clay minerals, the solutes moved rapidly through thick diameter cracks before the swelling of clay minerals. At the same time, preferential flow routes of a thinner diameter (earthworm burrows and root plant fissures) revealed a greater contribution to preferential flow when they became hydraulically active once thick diameter cracks start reducing due to clay minerals swelling.

In our study, intense irrigation or rainfall after a dry period which caused temporary flood conditions seemed to favor the preferential flow in clay-loam soils. Parameters such as initial water content, rainfall or irrigation intensity, as well as the timing and method of application of the solute in different soils are known to influence the preferential flow in different ways (Butters et al., 2000).

Irregular flow patterns and changes in the distribution of macroporosity from one irrigation to another have been reported by Ogden et al. (1999), resulting in high spatial variation in solute movement. However, Lee (1985) reported that preferential flow paths formed by earthworm activity persisted for 50–100 years in uncultivated soil, while Hagedorn and Bundt (2002) also reported high-persistence of preferential flow paths.

4. Conclusions

- Our data suggest that the main part of the corn-applied herbicides has been leached below the biological and adsorptively active surface soil, through preferential flow, within the first month after their application, whilst the greatest concentration of all parent compounds from all depths was detected in the first sampling when an irrigation or rainfall event follows shortly after pesticide application.
- Metolachlor, atrazine, DEA and DIA were detected in 99, 95, 91 and 67% of the total soil water samples, respectively.
- Alachlor was detected at concentrations greater than $0.1 \mu\text{g L}^{-1}$ up to 40 months after its application.
- Late metolachlor application, after the spring raining season, resulting in the reduction of its leaching.
- Metolachlor leaching period was more extended than that of atrazine.
- The concentration ratio ($C_{\text{met}}/C_{\text{atr}}$) in soil water samples from the 0–25 cm during the 1st and 2nd sampling were close to the concentration ration in their formulation. Then the ratio gradually increased up to two months after the mixture application. Afterward, the $C_{\text{met}}/C_{\text{atr}}$ ratio began to decrease again.
- By comparing the $C_{\text{al}}/C_{\text{atr}}$ a similar alachlor and atrazine adsorption behavior was observed. In addition, atrazine shown very high initial dissipation rate but four months after their application alachlor was less persistent than atrazine.
- Chromatographic flow, as shown to be dominant by the brilliant blue tracing, was restricted in the 10–15 cm bgs.
- Brilliant blue was detected below the depth of 150 cm within 24 h of its application following the local agricultural practice as a result of preferential flow caused by different types

of macropores (clay cracks, root channels, earthworm burrows).

- Late pesticide application, use of drip instead of sprinkler irrigation and delayed first irrigation seem to be the major management actions according to good agricultural practice that prevent pesticide leaching to groundwater in a semi-arid Mediterranean region.
- The obtained dataset represents an effective basis depicting the pesticide movement throughout soil profile. Raw dataset is also provided as Supplementary data in order to further be used for the calibration of pesticides leaching models.

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Appendix. Supplementary data

Supplementary data related to this article can be found online at [doi:10.1016/j.watres.2012.01.021](https://doi.org/10.1016/j.watres.2012.01.021).

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